MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, KEY CODE GHM

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.062 failures per million operating hours for class I or 0.119 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GHM.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/238-1 for class I and M28787/238-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

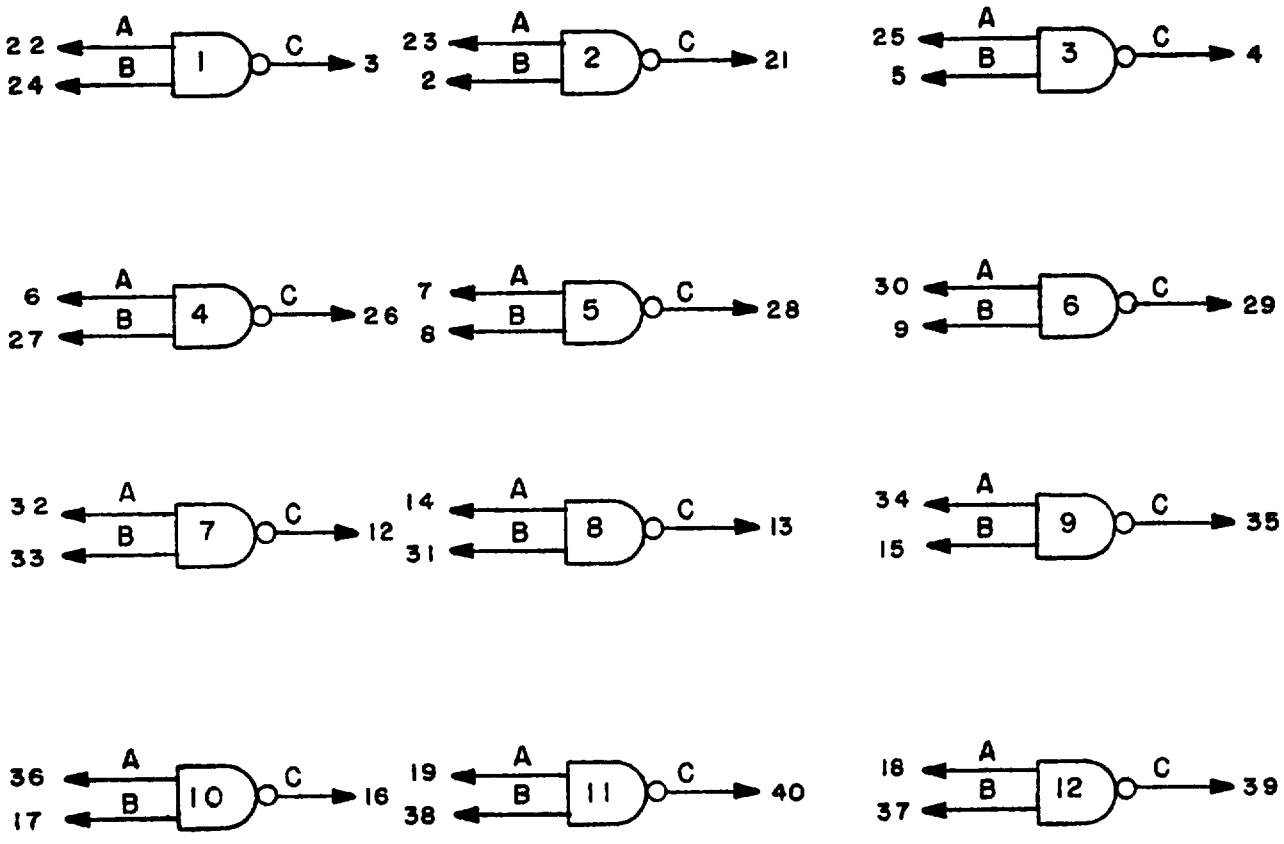
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

*4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. A = Gate inputs.
2. B = Gate inputs.
3. C = Gate outputs.

FIGURE 1. GHM module function diagram, NAND gate.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Frame Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1, B2	Unused pin isolation													
	Initial	IV	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C12	V_{OUT} (low loaded)	V	S	10	800	10	800	10	800	10	800	10	800	mV
D1-D24	Input breakdown current	VI	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	mA
E1-E24	Input clamp diode voltage	VII	S	-0.5	-1.5	-0.5	-1.5	-0.5	-1.5	-0.5	-1.5	-0.5	-1.5	V
F1-F12	Output voltage, low level	VIII	X	0	400	0	400	0	400	0	400	0	400	mV
G1-G36	Output leakage current, high level	IX	X	0	250	0	250	0	250	0	250	0	250	µA
H1-H24	Input current, low level	X	X	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	mA
J1-J24	Input current, high level	XI	X	0	30	0	40	0	40	0	40	0	40	µA
K1	Power supply current, low level	XII	S	1	162	1	162	1	162	1	162	1	162	mA
K2	Power supply current, high level	XII	S	1.0	25.5	1.0	25.5	1.0	25.5	1.0	25.5	1.0	25.5	mA
L1-L12	Transition time (t_{THL}), C_L=50 pF	XIII	S	2	20	2	25	2	25	2	25	2	25	ns
L13-L24	Transition time (t_{THL}), C_L=150 pF	XIII	Q	5	30	5	35	5	35	5	35	5	35	ns
M1-M24	Propagation delay time (t_{PLH}), C_L=50 pF	XIV	S	5	30	5	35	5	35	5	35	5	35	ns
M25-M48	Propagation delay time (t_{PLH}), C_L=150 pF	XIV	Q	5	40	5	45	5	45	5	45	5	45	ns
N1-N24	Propagation delay time (t_{PHL}), C_L=50 pF	XV	S	5	25	5	30	5	30	5	30	5	30	ns

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N25-N48	Propagation delay time (t_{PHL}), C_L=150 pF	XV	Q	5	35	5	40	5	40	5	40	5	40	ns
P1-P12	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
Q1-Q12	Output crosstalk, low level	XVII	Q	0	900	0	900	0	900	0	900	0	900	mV
R	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply filter capacitance	XIX	S	3.4	6.2	3.4	6.2	3.4	6.2	3.4	6.2	3.4	6.2	µF
T1-T24	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through S shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition S, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through S shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

*6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation			
Low level	891	891	mW
High level	140	140	mW
Input voltage			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V

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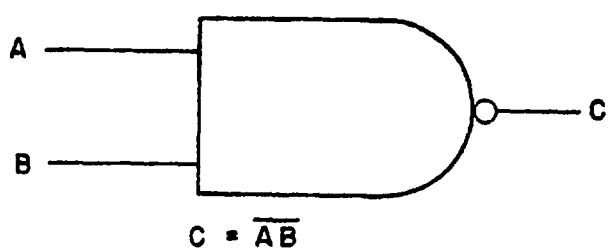
Input current			
Low level (V_{IL} = 0.4 V)	-1.6 max	-1.6 max	mA
High level (V_{IH} = 2.4 V)	40 max	40 max	µA
High level (V_{IH} = 5.5 V)	1.0 max	1.0 max	mA
Output voltage			
Low level	0.4 max	0.4 max	V
High level	5.5 max	5.5 max	V
Output drive current			
Low level (V_{OL} = 0.4 V)	-48 max	-48 max	mA
Output leakage current			
(V_{OUT} = 5.5 V)	250 max	250 max	µA
Propagation delay time			
t_{PLH} (C_L = 50 pF)	5 min, 35 max	5 min, 35 max	ns
(C_L = 150 pF)	5 min, 45 max	5 min, 45 max	ns
t_{PHL} (C_L = 50 pF)	5 min, 30 max	5 min, 30 max	ns
(C_L = 150 pF)	5 min, 40 max	5 min, 40 max	ns

*6.5 Module function. This module contains 12 independent 2-input open collector NAND gate/buffers as shown on figure 1.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic with pullup resistors provided external to the module.

TABLE II. Input to output relationship.

Input		Output
A	B	C
L	L	H
L	H	H
H	L	H
H	H	L



6.5.2 Wired-or. Wired-or is permitted with this open collector gate provided gate parameters are not exceeded.

6.5.2.1 Gate parameters. For the GHM module, the following parameters must not be exceeded for the wired-or function.

I_{OL}	48 mA maximum
V_{OH}	5.5 V maximum

6.5.2.2 Drive capability. The output drive capability of this combination is equal to the drive capability of an individual gate. The input loading is equal to the input loading of an individual gate.

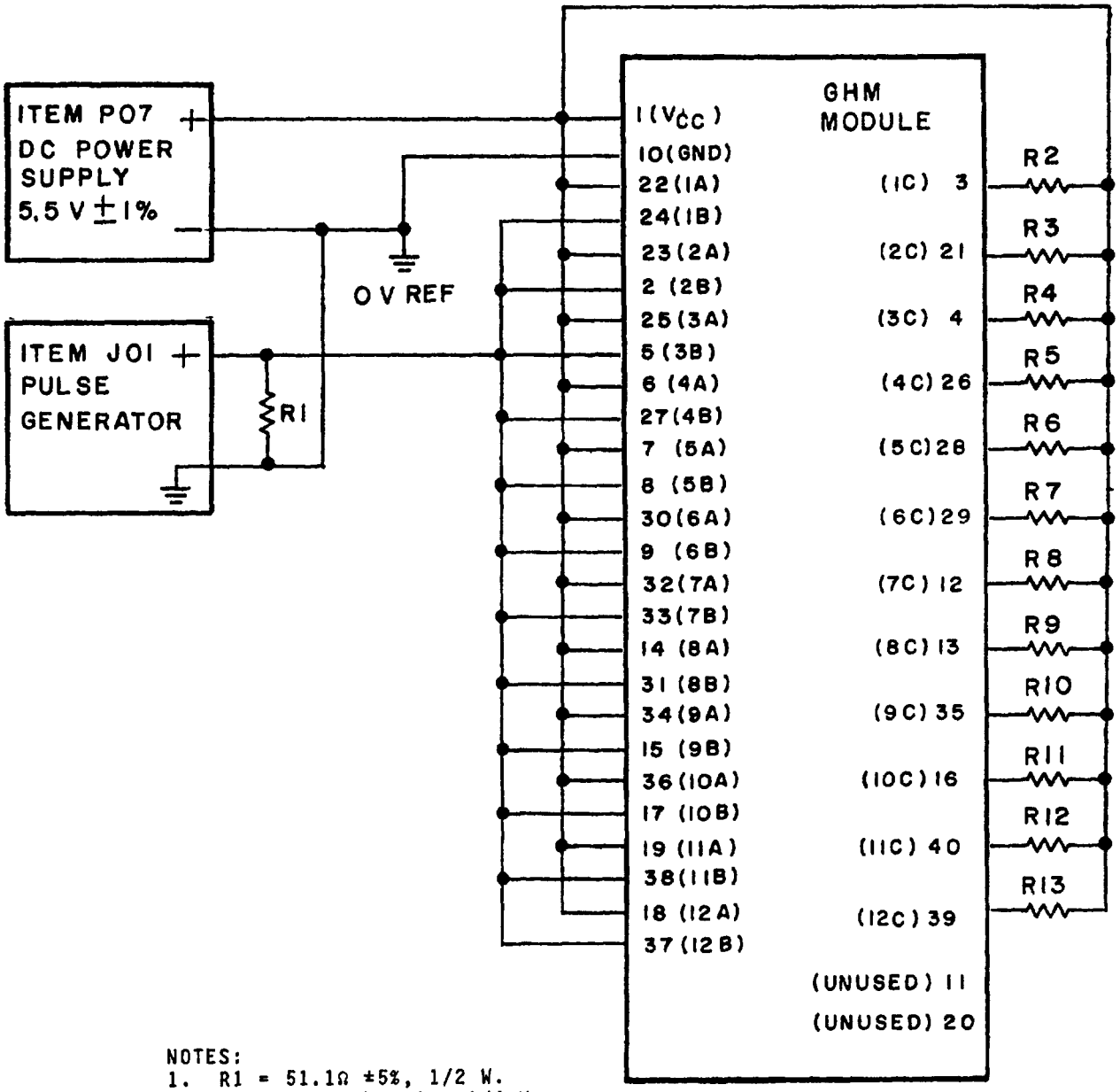
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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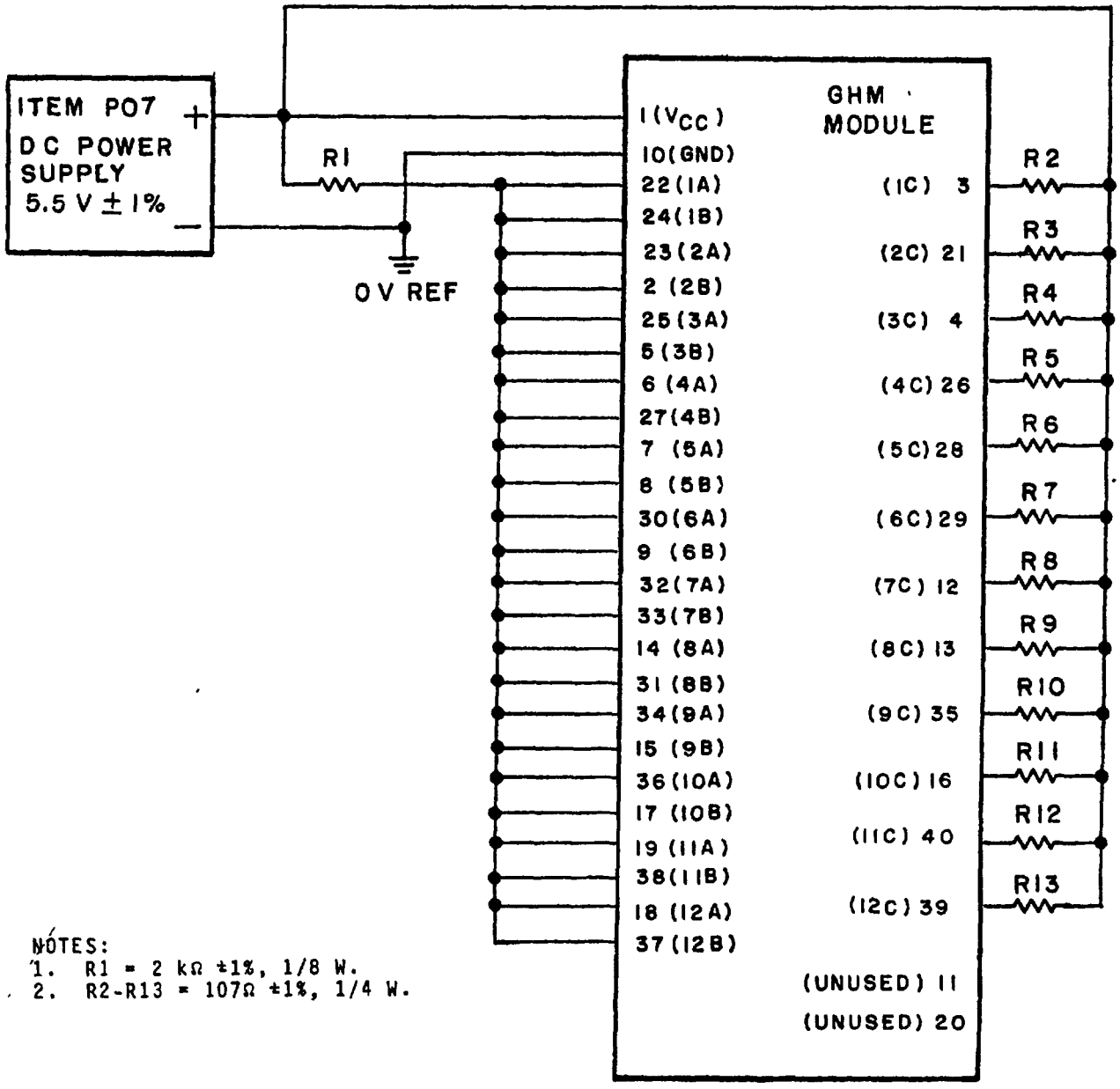
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2-R13 = 107Ω ±1%, 1/4 W.
3. Item J01 setup:
   PW = 500 ±50 ns.
   PRP = 1000 ±100 ns.
   PA = 3.0 ±0.1 V.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

* FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 2 kΩ ±1%, 1/8 W.
2. R2-R13 = 107Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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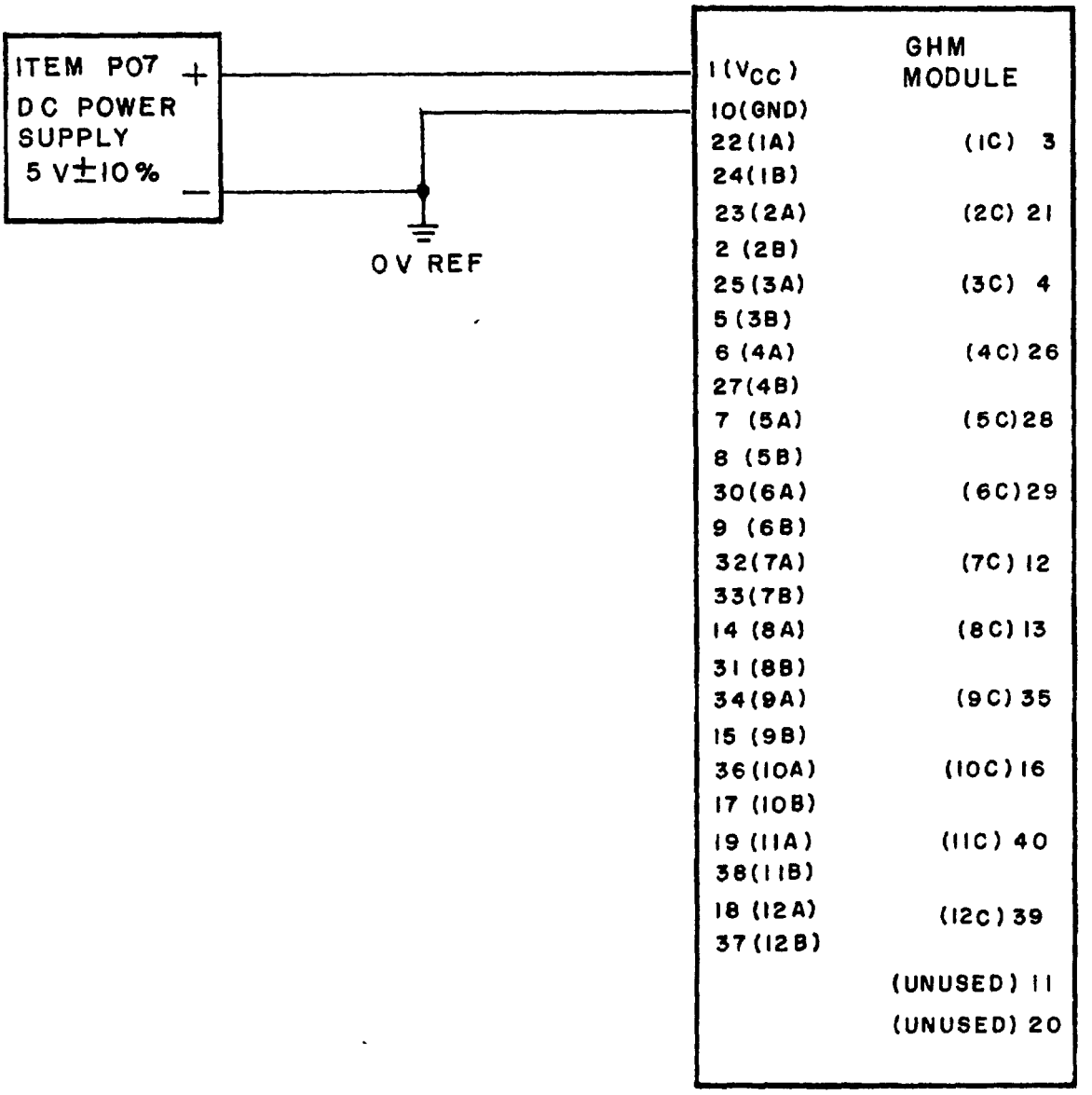


FIGURE 4. Vibration test circuit.

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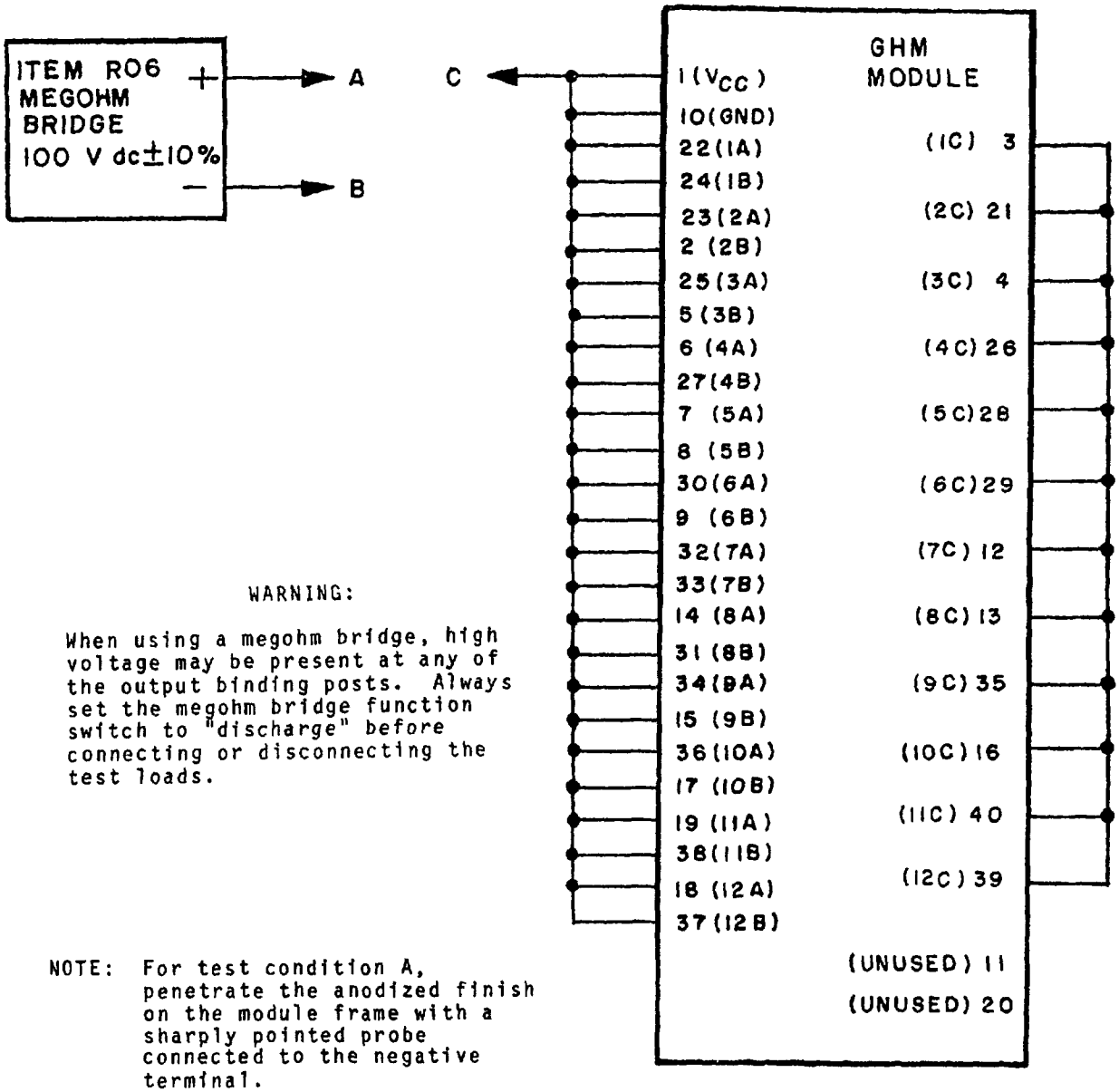
TABLE III. Frame isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Frame isolation	C,11,20	Frame	10	∞	MΩ	5

TABLE IV. Unused pin isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Unused pin isolation	C	11,20	10	∞	MΩ	5
B2	Unused pin isolation	11	20	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test loads.

NOTE: For test condition A, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

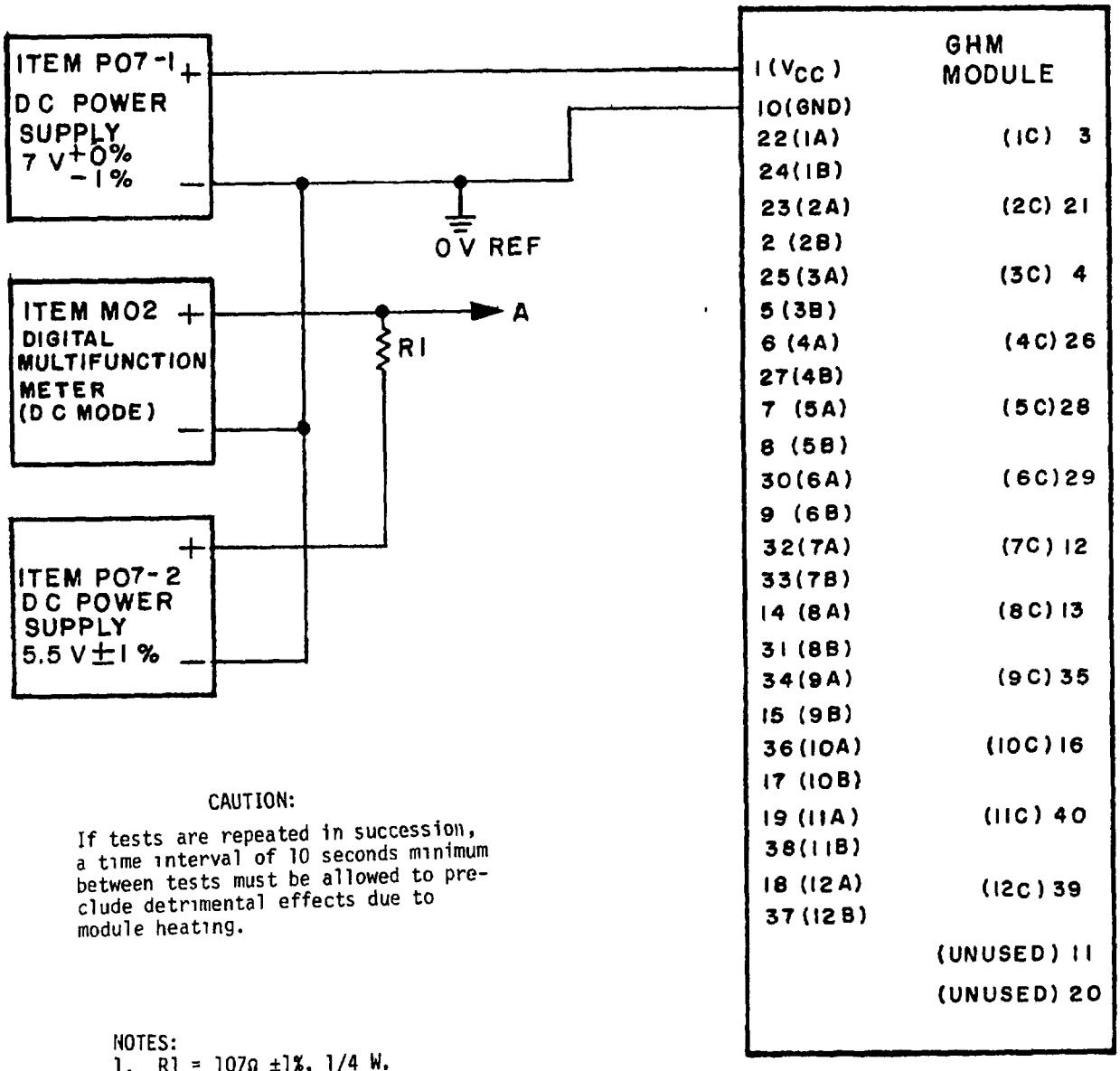
FIGURE 5. Isolation test circuit for tables III and IV.

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*TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C1	1C V_{OUT} (low loaded)	3	10	800	mV	6
C2	2C "	21	"	"	"	"
C3	3C "	4	"	"	"	"
C4	4C "	26	"	"	"	"
C5	5C "	28	"	"	"	"
C6	6C "	29	"	"	"	"
C7	7C "	12	"	"	"	"
C8	8C "	13	"	"	"	"
C9	9C "	35	"	"	"	"
C10	10C "	16	"	"	"	"
C11	11C "	40	"	"	"	"
C12	12C "	39	"	"	"	"

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CAUTION:

If tests are repeated in succession, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating.

NOTES:
1. R1 = 107Ω ±1%, 1/4 W.
2. After point A has been connected to the appropriate module pins, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1A Input breakdown current	22	24	0.01	1.00	mA	7
D2	1B "	24	22	"	"	"	"
D3	2A "	23	2	"	"	"	"
D4	2B "	2	23	"	"	"	"
D5	3A "	25	5	"	"	"	"
D6	3B "	5	25	"	"	"	"
D7	4A "	6	27	"	"	"	"
D8	4B "	27	6	"	"	"	"
D9	5A "	7	8	"	"	"	"
D10	5B "	8	7	"	"	"	"
D11	6A "	30	9	"	"	"	"
D12	6B "	9	30	"	"	"	"
D13	7A "	32	33	"	"	"	"
D14	7B "	33	32	"	"	"	"
D15	8A "	14	31	"	"	"	"
D16	8B "	31	14	"	"	"	"
D17	9A "	34	15	"	"	"	"
D18	9B "	15	34	"	"	"	"
D19	10A "	36	17	"	"	"	"
D20	10B "	17	36	"	"	"	"
D21	11A "	19	38	"	"	"	"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D22	11B Input breakdown current	38	19	0.01	1.00	mA	7
D23	12A "	18	37	"	"	"	"
D24	12B "	37	18	"	"	"	"

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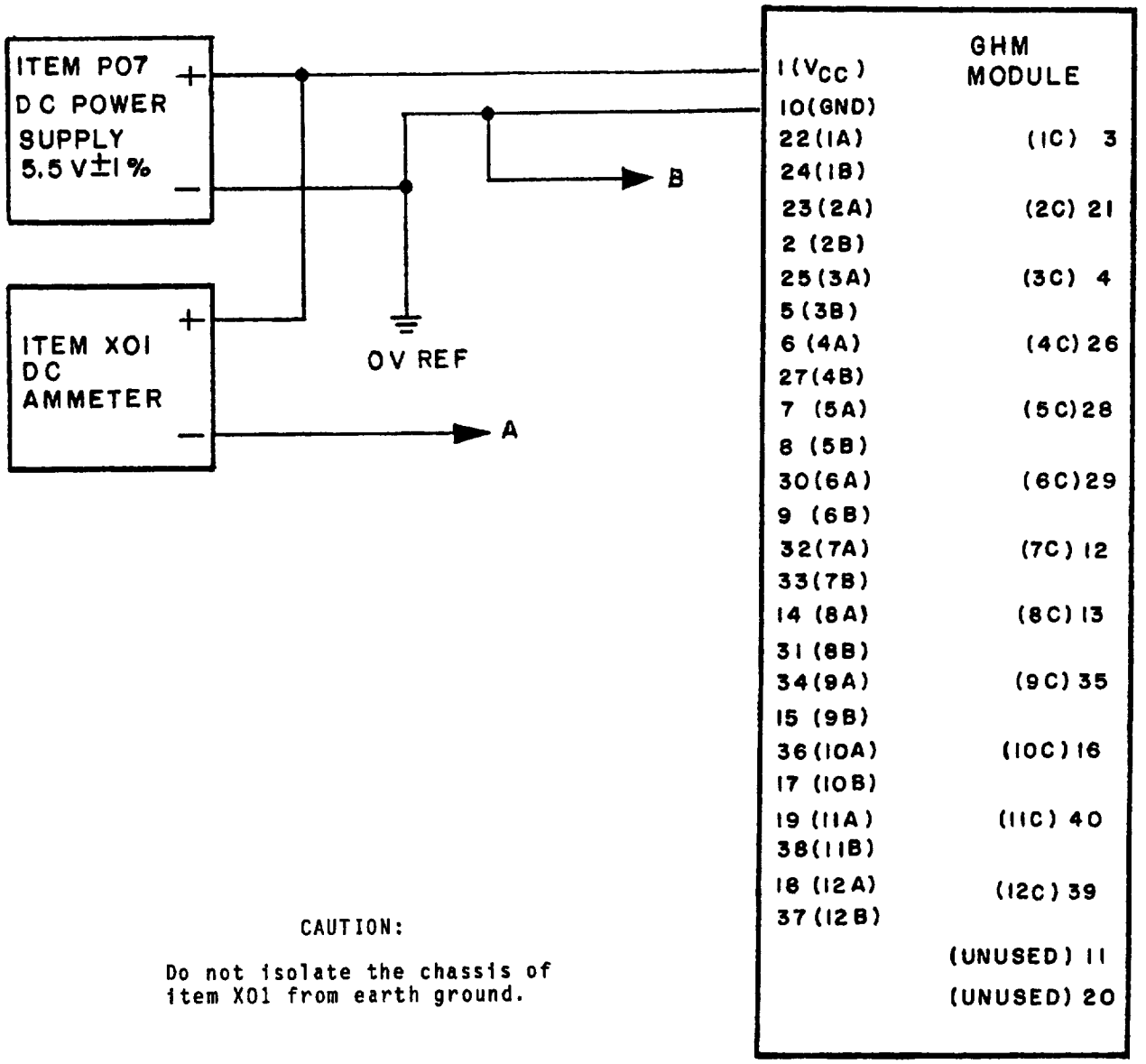


FIGURE 7. Input breakdown current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

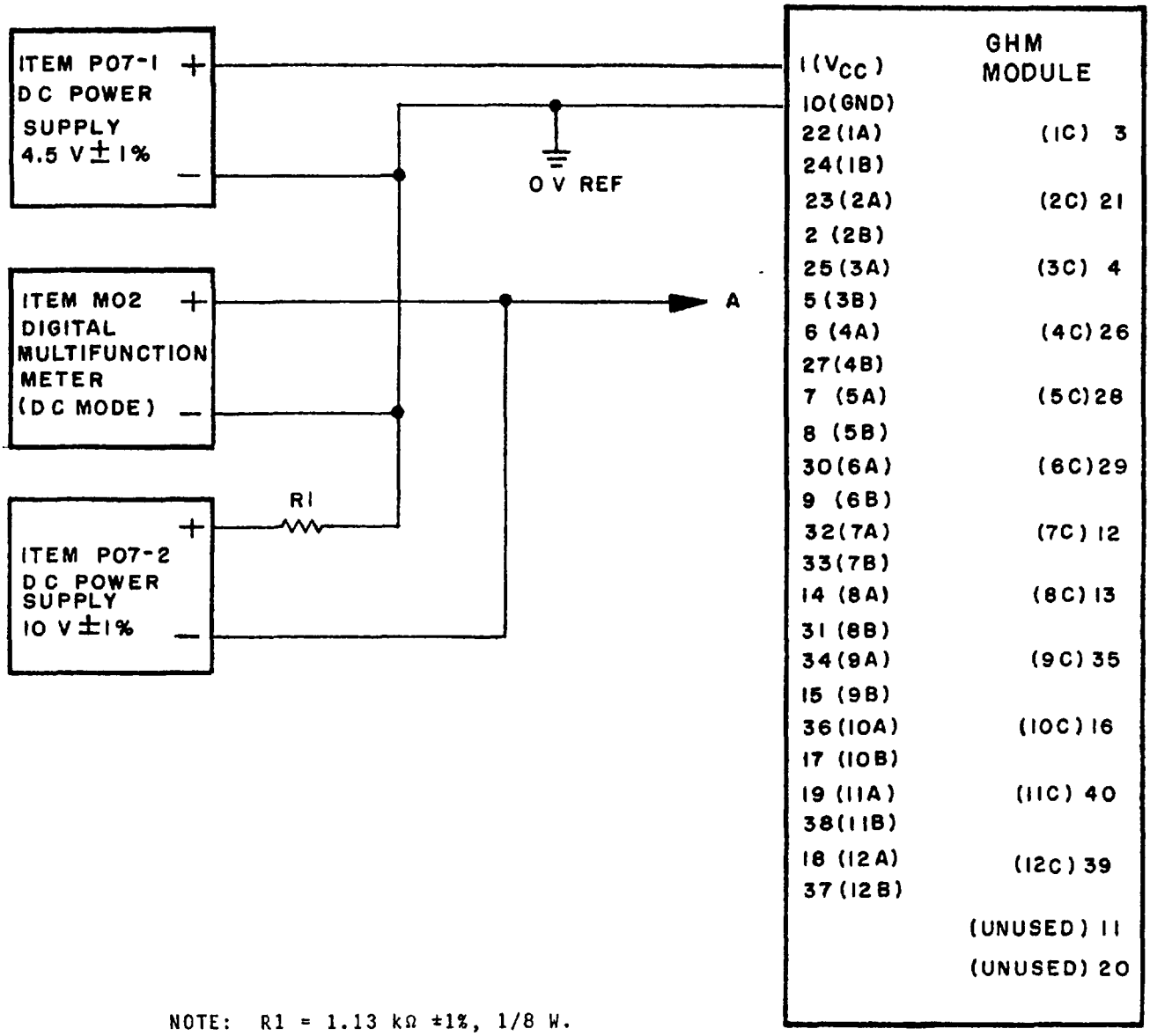
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1A Input clamp diode voltage	22	-0.5	-1.5	V	8
E2	1B "	24	"	"	"	"
E3	2A "	23	"	"	"	"
E4	2B "	2	"	"	"	"
E5	3A "	25	"	"	"	"
E6	3B "	5	"	"	"	"
E7	4A "	6	"	"	"	"
E8	4B "	27	"	"	"	"
E9	5A "	7	"	"	"	"
E10	5B "	8	"	"	"	"
E11	6A "	30	"	"	"	"
E12	6B "	9	"	"	"	"
E13	7A "	32	"	"	"	"
E14	7B "	33	"	"	"	"
E15	8A "	14	"	"	"	"
E16	8B "	31	"	"	"	"
E17	9A "	34	"	"	"	"
E18	9B "	15	"	"	"	"
E19	10A "	36	"	"	"	"
E20	10B "	17	"	"	"	"
E21	11A "	19	"	"	"	"
E22	11B "	38	"	"	"	"

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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E23	12A Input clamp diode voltage	18	-0.5	-1.5	V	8
E24	12B Input clamp diode voltage	37	-0.5	-1.5	V	8

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NOTE: R1 = 1.13 kΩ ±1%, 1/8 W.

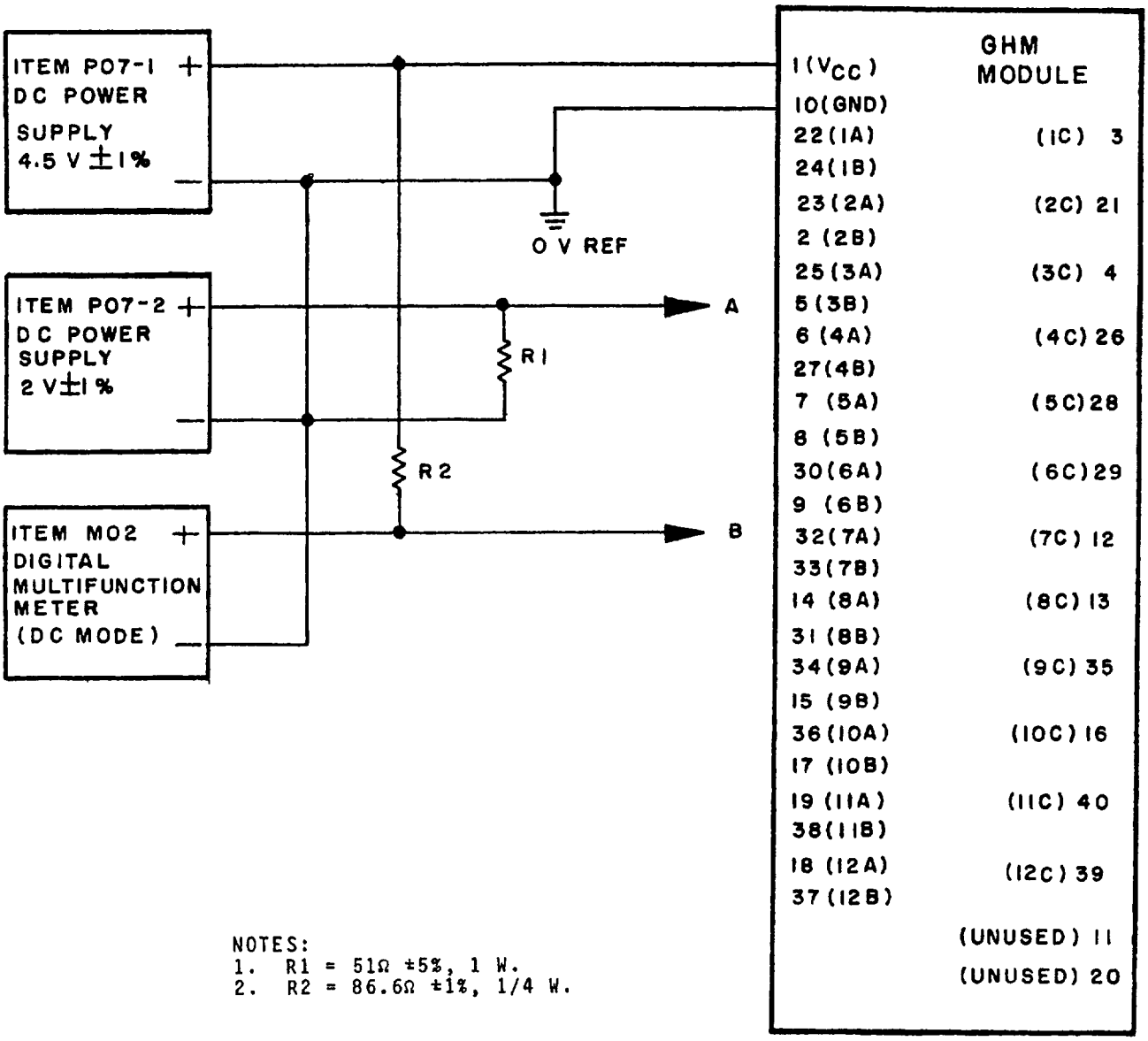
* FIGURE 8. Input clamp diode voltage test circuit for table VII.

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*TABLE VIII. Output voltage (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	1C Output voltage, low level	22,24	3	0	400	mV	9
F2	2C "	23,2	21	"	"	"	"
F3	3C "	25,5	4	"	"	"	"
F4	4C "	6,27	26	"	"	"	"
F5	5C "	7,8	28	"	"	"	"
F6	6C "	30,9	29	"	"	"	"
F7	7C "	32,33	12	"	"	"	"
F8	8C "	14,31	13	"	"	"	"
F9	9C "	34,15	35	"	"	"	"
F10	10C "	36,17	16	"	"	"	"
F11	11C "	19,38	40	"	"	"	"
F12	12C "	18,37	39	"	"	"	"

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 86.6Ω ±1%, 1/4 W.

* FIGURE 9. Output voltage (low-level) test circuit for table VIII.

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TABLE IX. Output leakage current (high-level) test.

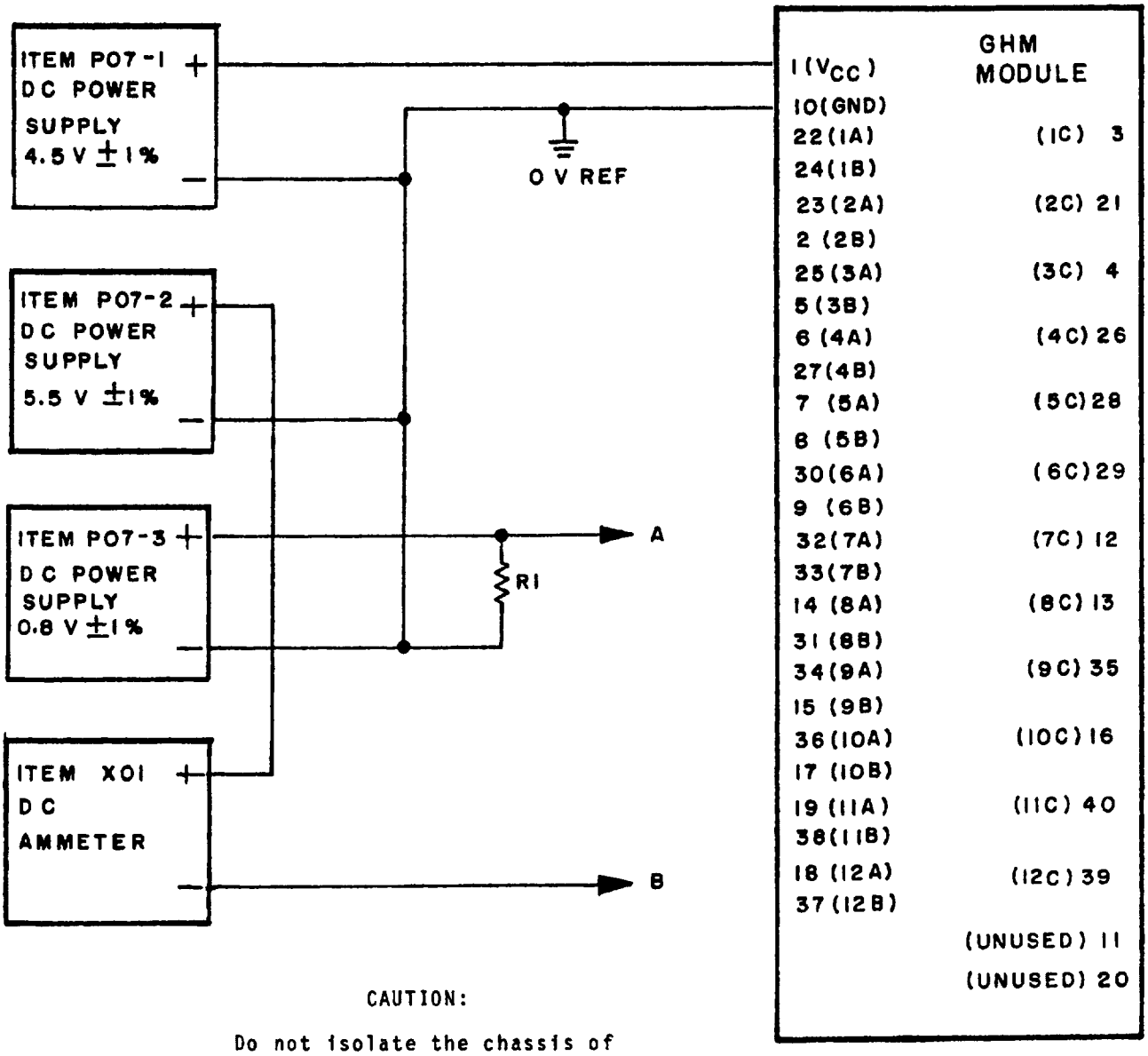
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	1C Output leakage current, high level	22	3	0	250	µA	10
G2	1C "	24	3	"	"	"	"
G3	2C "	23	21	"	"	"	"
G4	2C "	2	21	"	"	"	"
G5	3C "	25	4	"	"	"	"
G6	3C "	5	4	"	"	"	"
G7	4C "	6	26	"	"	"	"
G8	4C "	27	26	"	"	"	"
G9	5C "	7	28	"	"	"	"
G10	5C "	8	28	"	"	"	"
G11	6C "	30	29	"	"	"	"
G12	6C "	9	29	"	"	"	"
G13	7C "	32	12	"	"	"	"
G14	7C "	33	12	"	"	"	"
G15	8C "	14	13	"	"	"	"
G16	8C "	31	13	"	"	"	"
G17	9C "	34	35	"	"	"	"
G18	9C "	15	35	"	"	"	"
G19	10C "	36	16	"	"	"	"
G20	10C "	17	16	"	"	"	"
G21	11C "	19	40	"	"	"	"
G22	11C "	38	40	"	"	"	"

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TABLE IX. Output leakage current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G23	12C Output leakage current, high level	18	39	0	250	µA	10
G24	12C "	37	39	"	"	"	"
G25	1C "	22,24	3	"	"	"	"
G26	2C "	23,2	21	"	"	"	"
G27	3C "	25,5	4	"	"	"	"
G28	4C "	6,27	26	"	"	"	"
G29	5C "	7,8	28	"	"	"	"
G30	6C "	30,9	29	"	"	"	"
G31	7C "	32,33	12	"	"	"	"
G32	8C "	14,31	13	"	"	"	"
G33	9C "	34,15	35	"	"	"	"
G34	10C "	36,17	16	"	"	"	"
G35	11C "	19,38	40	"	"	"	"
G36	12C "	18,37	39	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 10. Output leakage current test circuit for table IX.

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TABLE X. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1A Input current, low level	Open	24	Open	-0.40	-1.73	mA	11
H2	1B "	"	22	"	"	"	"	"
H3	2A "	"	2	"	"	"	"	"
H4	2B "	"	23	"	"	"	"	"
H5	3A "	"	5	"	"	"	"	"
H6	3B "	"	25	"	"	"	"	"
H7	4A "	"	27	"	"	"	"	"
H8	4B "	"	6	"	"	"	"	"
H9	5A "	"	8	"	"	"	"	"
H10	5B "	"	7	"	"	"	"	"
H11	6A "	"	9	"	"	"	"	"
H12	6B "	"	30	"	"	"	"	"
H13	7A "	"	33	"	"	"	"	"
H14	7B "	"	32	"	"	"	"	"
H15	8A "	"	31	"	"	"	"	"
H16	8B "	"	14	"	"	"	"	"
H17	9A "	"	15	"	"	"	"	"
H18	9B "	"	34	"	"	"	"	"
H19	10A "	"	17	"	"	"	"	"
H20	10B "	"	36	"	"	"	"	"
H21	11A "	"	38	"	"	"	"	"
H22	11B "	"	19	"	"	"	"	"

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TABLE X. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H23	12A Input current, low level	Open	37	Open	-0.40	-1.73	mA	11
H24	12B Input current, low level	Open	18	Open	-0.40	-1.73	mA	11

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TABLE XI. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1A Input current, high level	22	Open	24	0	30	µA	11
J2	1B "	24	"	22	"	"	"	"
J3	2A "	23	"	2	"	"	"	"
J4	2B "	2	"	23	"	"	"	"
J5	3A "	25	"	5	"	"	"	"
J6	3B "	5	"	25	"	"	"	"
J7	4A "	6	"	27	"	"	"	"
J8	4B "	27	"	6	"	"	"	"
J9	5A "	7	"	8	"	"	"	"
J10	5B "	8	"	7	"	"	"	"
J11	6A "	30	"	9	"	"	"	"
J12	6B "	9	"	30	"	"	"	"
J13	7A "	32	"	33	"	"	"	"
J14	7B "	33	"	32	"	"	"	"
J15	8A "	14	"	31	"	"	"	"
J16	8B "	31	"	14	"	"	"	"
J17	9A "	34	"	15	"	"	"	"
J18	9B "	15	"	34	"	"	"	"
J19	10A "	36	"	17	"	"	"	"
J20	10B "	17	"	36	"	"	"	"
J21	11A "	19	"	38	"	"	"	"
J22	11B "	38	"	19	"	"	"	"

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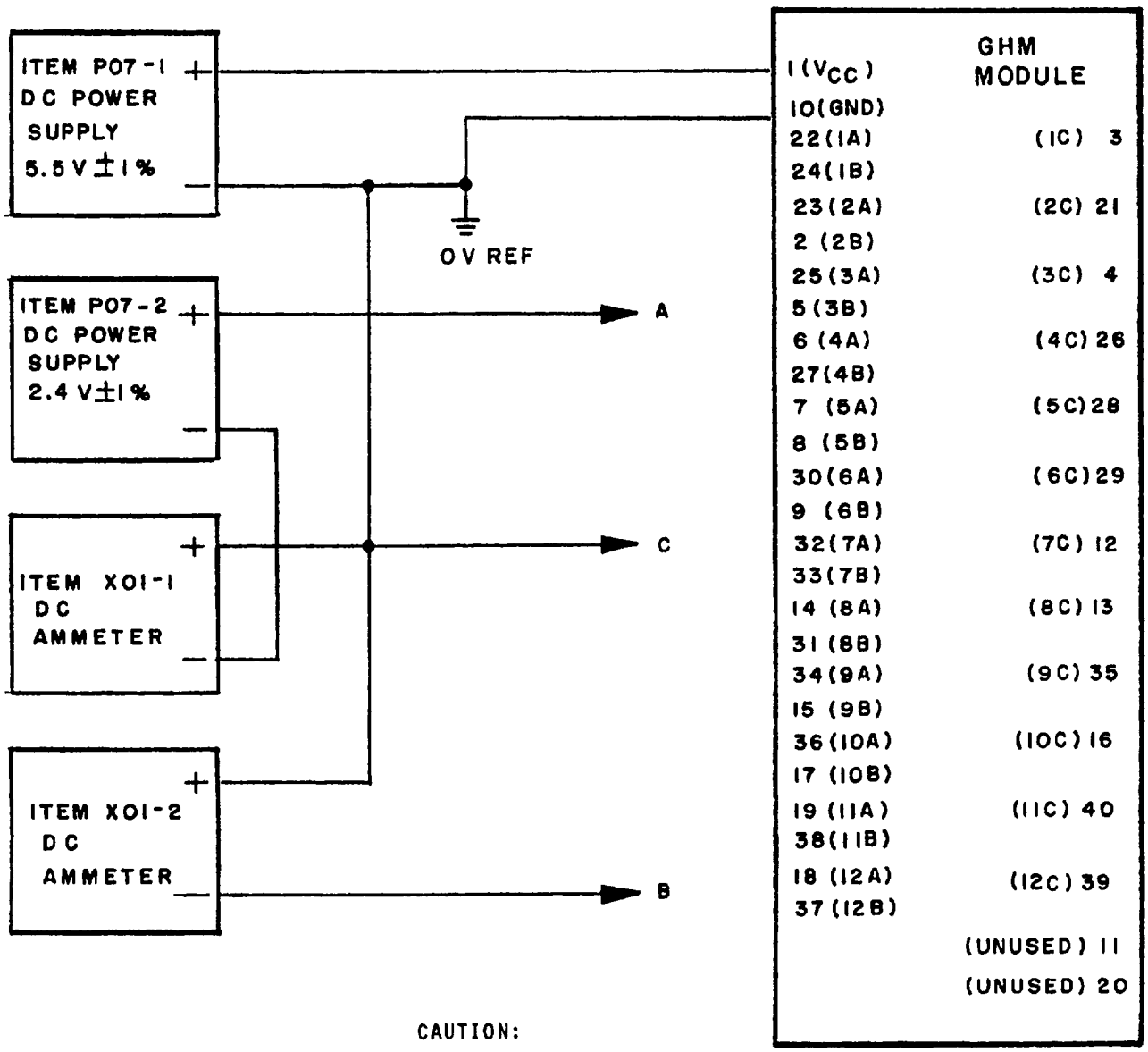
TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J23	12A Input current, high level	18	Open	37	0	30	µA	11
J24	12B Input current, high level	37	Open	18	0	30	µA	11

TABLE XII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Power supply current, low level	Open	Open	1	162	mA	12
K2	Power supply current, high level	B	A	1.0	25.5	mA	12

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.

* FIGURE 11. Input current (low-level and high-level) test circuit for tables X and XI.

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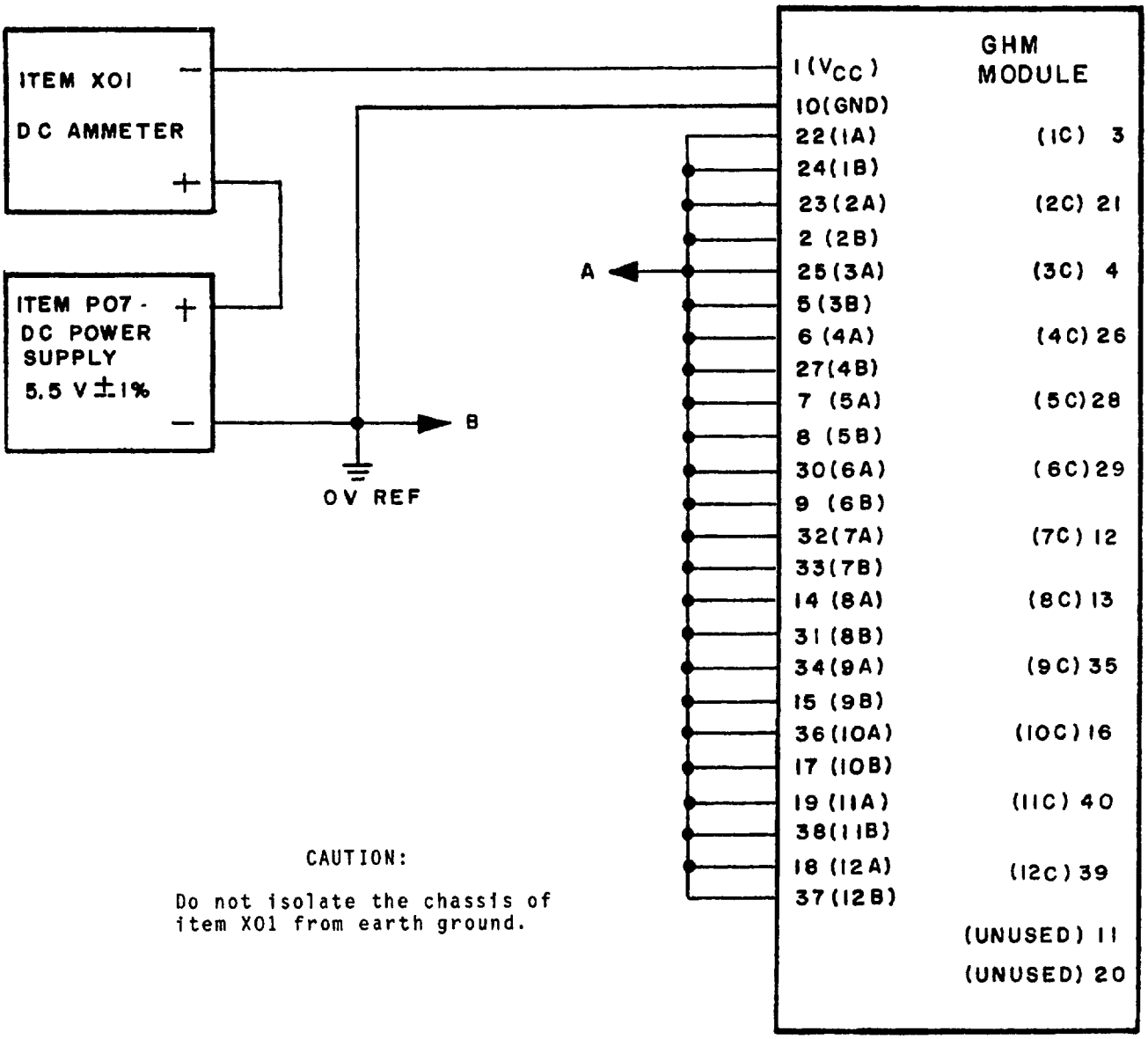


FIGURE 12. Power supply current (low-level and high-level) test circuit for table XII.

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	C1 Transition time (t_{THL}), C_L = 50 pF	22	24	3	2	20	ns	13,14
L2	C2 "	23	2	21	"	"	"	"
L3	C3 "	25	5	4	"	"	"	"
L4	C4 "	6	27	26	"	"	"	"
L5	C5 "	7	8	28	"	"	"	"
L6	C6 "	30	9	29	"	"	"	"
L7	C7 "	32	33	12	"	"	"	"
L8	C8 "	14	31	13	"	"	"	"
L9	C9 "	34	15	35	"	"	"	"
L10	C10 "	36	17	16	"	"	"	"
L11	C11 "	19	38	40	"	"	"	"
L12	C12 "	18	37	39	"	"	"	"
L13	C1 Transition time (t_{THL}), C_L = 150 pF	22	24	3	5	30	"	"
L14	C2 "	23	2	21	"	"	"	"
L15	C3 "	25	5	4	"	"	"	"
L16	C4 "	6	27	26	"	"	"	"
L17	C5 "	7	8	28	"	"	"	"
L18	C6 "	30	9	29	"	"	"	"
L19	C7 "	32	33	12	"	"	"	"
L20	C8 "	14	31	13	"	"	"	"

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TABLE XIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L21	C9 Transition time (t_{THL}), C_L = 150 pF	34	15	35	5	30	ns	13,14
L22	C10 "	36	17	16	"	"	"	"
L23	C11 "	19	38	40	"	"	"	"
L24	C12 "	18	37	39	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	C1 Propagation delay time (t_{PLH}), C_L = 50 pF	22	24	3	5	30	ns	13,14
M2	C2 "	23	2	21	"	"	"	"
M3	C3 "	25	5	4	"	"	"	"
M4	C4 "	6	27	26	"	"	"	"
M5	C5 "	7	8	28	"	"	"	"
M6	C6 "	30	9	29	"	"	"	"
M7	C7 "	32	33	12	"	"	"	"
M8	C8 "	14	31	13	"	"	"	"
M9	C9 "	34	15	35	"	"	"	"
M10	C10 "	36	17	16	"	"	"	"
M11	C11 "	19	38	40	"	"	"	"
M12	C12 "	18	37	39	"	"	"	"
M13	C1 "	24	22	3	"	"	"	"
M14	C2 "	2	23	21	"	"	"	"
M15	C3 "	5	25	4	"	"	"	"
M16	C4 "	27	6	26	"	"	"	"
M17	C5 "	8	7	28	"	"	"	"
M18	C6 "	9	30	29	"	"	"	"
M19	C7 "	33	32	12	"	"	"	"
M20	C8 "	31	14	13	"	"	"	"
M21	C9 "	15	34	35	"	"	"	"
M22	C10 "	17	36	16	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M23	C11 Propagation delay time (t_{PLH}), C_L = 50 pF	38	19	40	5	30	ns	13,14
M24	C12 Propagation delay time (t_{PLH}), C_L = 50 pF	37	18	39	"	30	"	"
M25	C1 Propagation delay time (t_{PLH}), C_L = 150 pF	22	24	3	"	40	"	"
M26	C2 "	23	2	21	"	"	"	"
M27	C3 "	25	5	4	"	"	"	"
M28	C4 "	6	27	26	"	"	"	"
M29	C5 "	7	8	28	"	"	"	"
M30	C6 "	30	9	29	"	"	"	"
M31	C7 "	32	33	12	"	"	"	"
M32	C8 "	14	31	13	"	"	"	"
M33	C9 "	34	15	35	"	"	"	"
M34	C10 "	36	17	16	"	"	"	"
M35	C11 "	19	38	40	"	"	"	"
M36	C12 "	18	37	39	"	"	"	"
M37	C1 "	24	22	3	"	"	"	"
M38	C2 "	2	23	21	"	"	"	"
M39	C3 "	5	25	4	"	"	"	"
M40	C4 "	27	6	26	"	"	"	"
M41	C5 "	8	7	28	"	"	"	"
M42	C6 "	9	30	29	"	"	"	"
M43	C7 "	33	32	12	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M44	C8 Propagation delay time (t_{PLH}), C_L = 150 pF	31	14	13	5	40	ns	13,14
M45	C9 "	15	34	35	"	"	"	"
M46	C10 "	17	36	16	"	"	"	"
M47	C11 "	38	19	40	"	"	"	"
M48	C12 "	37	18	39	"	"	"	"

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	C1 Propagation delay time (t_{PHL}), C_L = 50 pF	22	24	3	5	25	ns	13,14
N2	C2 "	23	2	21	"	"	"	"
N3	C3 "	25	5	4	"	"	"	"
N4	C4 "	6	27	26	"	"	"	"
N5	C5 "	7	8	28	"	"	"	"
N6	C6 "	30	9	29	"	"	"	"
N7	C7 "	32	33	12	"	"	"	"
N8	C8 "	14	31	13	"	"	"	"
N9	C9 "	34	15	35	"	"	"	"
N10	C10 "	36	17	16	"	"	"	"
N11	C11 "	19	38	40	"	"	"	"
N12	C12 "	18	37	39	"	"	"	"
N13	C1 "	24	22	3	"	"	"	"
N14	C2 "	2	23	21	"	"	"	"
N15	C3 "	5	25	4	"	"	"	"
N16	C4 "	27	6	26	"	"	"	"
N17	C5 "	8	7	28	"	"	"	"
N18	C6 "	9	30	29	"	"	"	"
N19	C7 "	33	32	12	"	"	"	"
N20	C8 "	31	14	13	"	"	"	"
N21	C9 "	15	34	35	"	"	"	"
N22	C10 "	17	36	16	"	"	"	"

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

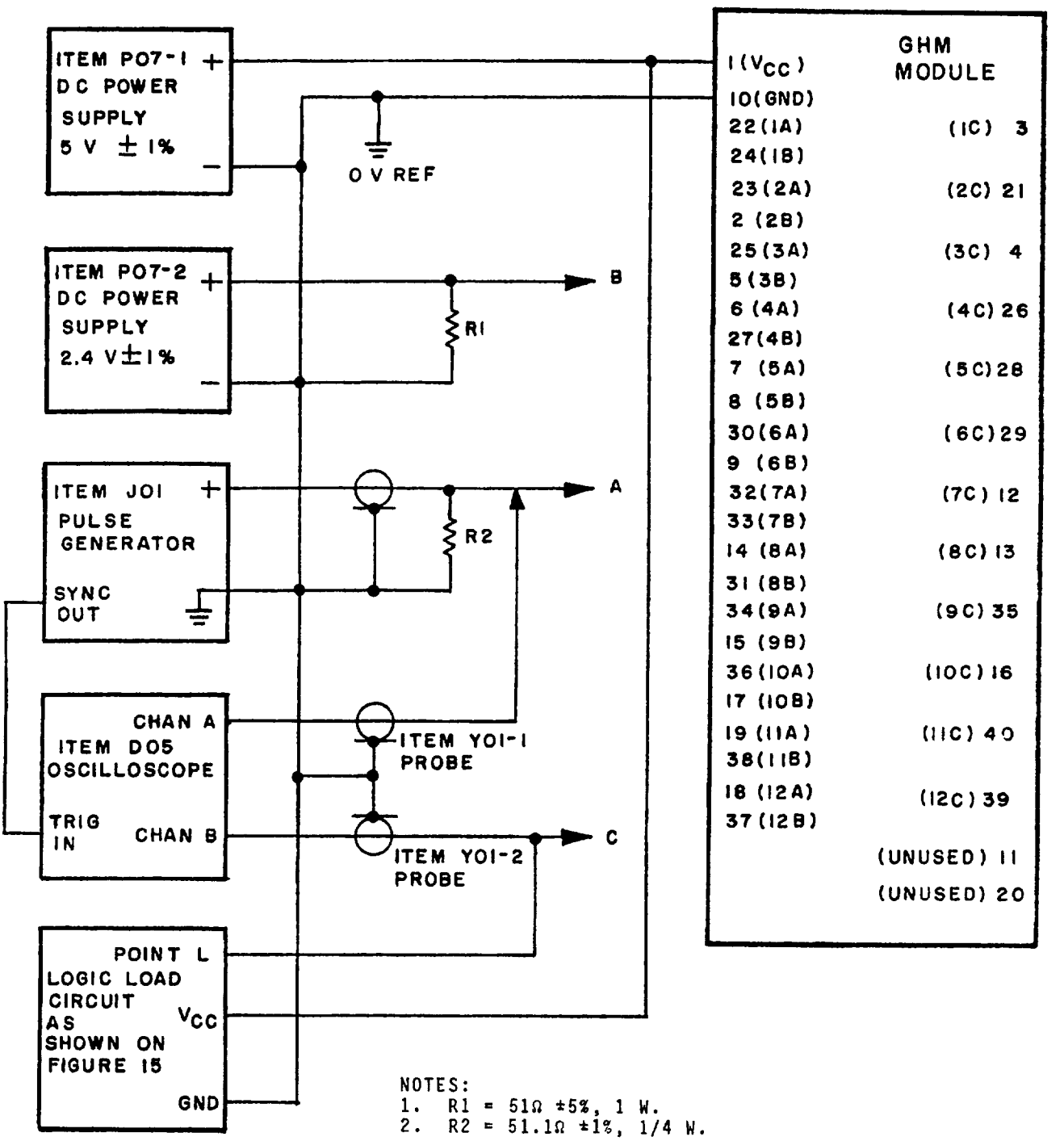
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N23	C11 Propagation delay time (t_{PHL}), C_L = 50 pF	38	19	40	5	25	ns	13,14
N24	C12 Propagation delay time (t_{PHL}), C_L = 50 pF	37	18	39	"	25	"	"
N25	C1 Propagation delay time (t_{PHL}), C_L = 150 pF	22	24	3	"	35	"	"
N26	C2 "	23	2	21	"	"	"	"
N27	C3 "	25	5	4	"	"	"	"
N28	C4 "	6	27	26	"	"	"	"
N29	C5 "	7	8	28	"	"	"	"
N30	C6 "	30	9	29	"	"	"	"
N31	C7 "	32	33	12	"	"	"	"
N32	C8 "	14	31	13	"	"	"	"
N33	C9 "	34	15	35	"	"	"	"
N34	C10 "	36	17	16	"	"	"	"
N35	C11 "	19	38	40	"	"	"	"
N36	C12 "	18	37	39	"	"	"	"
N37	C1 "	24	22	3	"	"	"	"
N38	C2 "	2	23	21	"	"	"	"
N39	C3 "	5	25	4	"	"	"	"
N40	C4 "	27	6	26	"	"	"	"
N41	C5 "	8	7	28	"	"	"	"
N42	C6 "	9	30	29	"	"	"	"
N43	C7 "	33	32	12	"	"	"	"

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Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N44	C8 Propagation delay time (t_{PHL}), C_L = 150 pF	31	14	13	5	35	ns	13,14
N45	C9 "	15	34	35	"	"	"	"
N46	C10 "	17	36	16	"	"	"	"
N47	C11 "	38	19	40	"	"	"	"
N48	C12 "	37	18	39	"	"	"	"

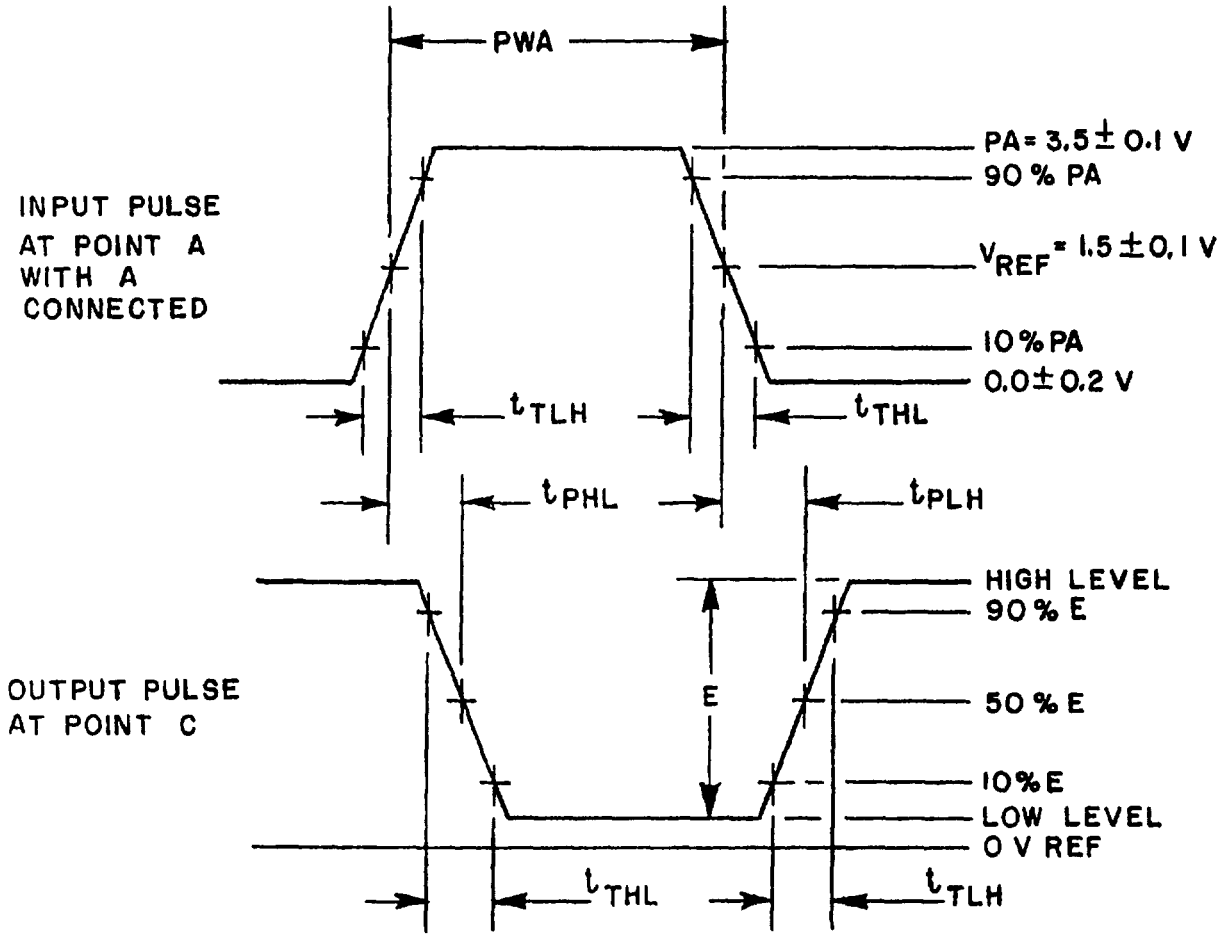
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 51.1Ω ±1%, 1/4 W.

* FIGURE 13. Transition time (t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIII, XIV, and XV.

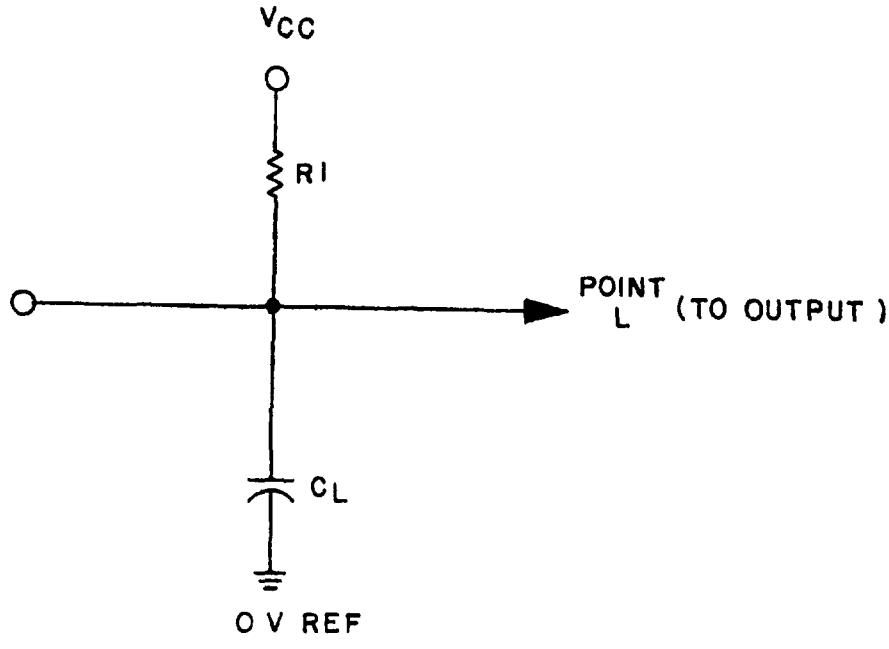
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NOTES:
1. Input pulse parameters:
   PWA = 100 ns ±10%.
   $t_{TLH} = t_{THL}$ = 10 ns ±10%.
   PRR = 5 MHz ±10%.
2. All voltages are referenced to ground.
3. Verify that the output has a high level ≥ 2.4 V and a low level ≤ 0.4 V before making measurements.

* FIGURE 14. Transition time (t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIII, XIV, and XV.

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NOTES:
1. R1 = 97.6Ω ±5%, 1/4 W.
2. For test conditions L1-L12, M1-M24, N1-N24, P1-P12, and Q1-Q12, C_L = 50 ±5 pF (including probe and fixture capacitance).
3. For test conditions L13-L24, M25-M48, N25-N48, C_L = 150 ±5 pF (including probe and fixture capacitance).
4. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

FIGURE 15. Logic load circuit.

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TABLE XVI. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1C Output crosstalk, high level	2,5,27,8,9,33,31,15,17,38,37	22,24	23,25,6,7,30,32,14,34,36,18,19	3	0	400	mV	16,17
P2	2C "	24,5,27,8,9,33,31,15,17,38,37	23,2	22,25,6,7,30,32,14,34,36,19,18	21	"	"	"	"
P3	3C "	24,2,27,8,9,33,31,15,17,38,37	25,5	22,23,6,7,30,32,14,34,36,19,18	4	"	"	"	"
P4	4C "	24,2,5,8,9,33,31,15,17,38,37	6,27	22,23,25,7,30,32,14,34,36,19,18	26	"	"	"	"
P5	5C "	24,2,5,27,9,33,31,15,17,38,37	7,8	22,23,25,6,30,32,14,34,36,19,18	28	"	"	"	"
P6	6C "	24,2,5,27,8,33,31,15,17,38,37	30,9	22,23,25,6,7,32,14,34,36,18,19	29	"	"	"	"
P7	7C "	24,2,5,27,8,9,31,15,17,38,37	32,33	22,23,25,6,7,30,14,34,36,19,18	12	"	"	"	"
P8	8C "	24,2,5,27,8,9,33,15,17,38,37	14,31	22,23,25,6,7,30,32,34,36,19,18	13	"	"	"	"
P9	9C "	24,2,5,27,8,9,33,31,17,38,37	34,15	22,23,25,6,7,30,32,14,36,19,18	35	"	"	"	"
P10	10C "	24,2,5,27,8,9,33,31,15,38,37	36,17	22,23,25,6,7,30,32,14,34,19,18	16	"	"	"	"

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TABLE XVI. Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P11	11C Output crosstalk, high level	24,2,5,27, 8,9,33,31, 15,17,37	19,38	22,23,25,6, 7,30,32,14, 34,36,18	40	0	400	mV	16,17
P12	12C Output crosstalk, high level	24,2,5,27, 8,9,33,31, 15,17,38	18,37	22,23,25,6, 7,30,32,14, 34,36,19	39	0	400	mV	16,17

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TABLE XVII. Output crosstalk (low-level) test.

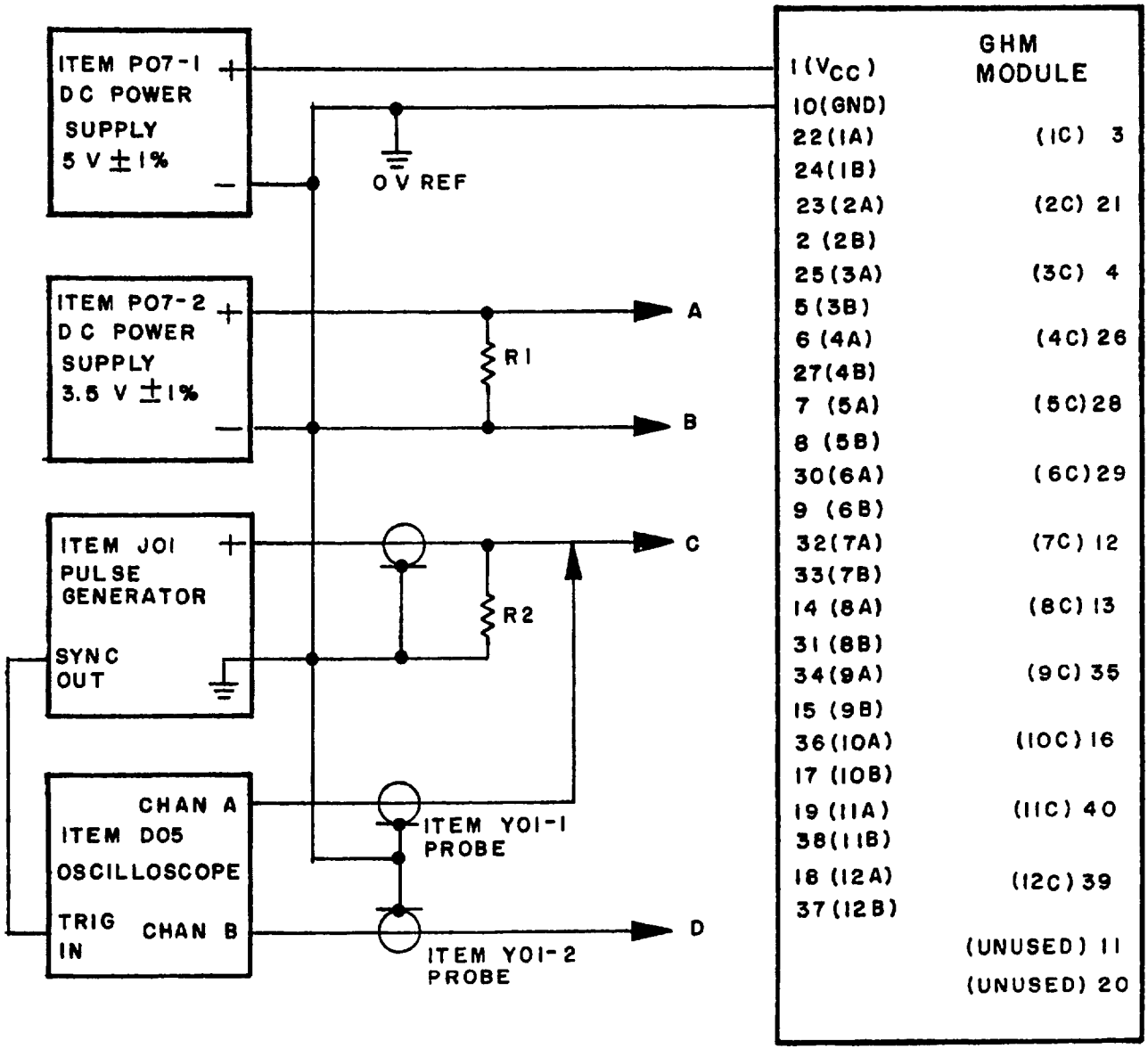
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1C Output crosstalk, low level	22,24,2,5, 27,8,9,33, 31,15,17, 38,37	Open	23,25,6,7, 30,32,14,34, 36,19,18	3	0	900	mV	16,17
Q2	2C "	24,23,2,5, 27,8,9,33, 31,15,17, 38,37	"	22,25,6,7, 30,32,14,34, 36,19,18	21	"	"	"	"
Q3	3C "	24,2,25,5, 27,8,9,33, 31,15,17, 38,37	"	22,23,6,7, 30,32,14,34, 36,19,18	4	"	"	"	"
Q4	4C "	24,2,5,6, 27,8,9,33, 31,15,17, 38,37	"	22,23,25,7, 30,32,14,34, 36,19,18	26	"	"	"	"
Q5	5C "	24,2,5,27, 7,8,9,33, 31,15,17, 38,37	"	22,23,25,6, 30,32,14,34, 36,19,18	28	"	"	"	"
Q6	6C "	24,2,5,27, 8,30,9,33, 31,15,17, 38,37	"	22,23,25,6, 7,32,14,34, 36,19,18	29	"	"	"	"
Q7	7C "	24,2,5,27, 8,9,32,33, 31,15,17, 38,37	"	22,23,25,6, 7,30,14,34, 36,19,18	12	"	"	"	"
Q8	8C "	24,2,5,27, 8,9,33,14, 31,15,17, 38,37	"	22,23,25,6, 7,30,32,34, 36,19,18	13	"	"	"	"

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TABLE XVII. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q9	9C Output crosstalk, low level	24,2,5,27, 8,9,33,31, 34,15,17, 38,37	Open	22,23,25,6, 7,30,32,14, 36,19,18	35	0	900	mV	16,17
Q10	10C "	24,2,5,27, 8,9,33,31, 15,36,17, 38,37	"	22,23,25,6, 7,30,32,14, 34,19,18	16	"	"	"	"
Q11	11C "	24,2,5,27, 8,9,33,31, 15,17,19, 38,37	"	22,23,25,6, 7,30,32,14, 34,36,18	40	"	"	"	"
Q12	12C "	24,2,5,27, 8,9,33,31, 15,17,38, 18,37	"	22,23,25,6, 7,30,32,14, 34,36,19	39	"	"	"	"

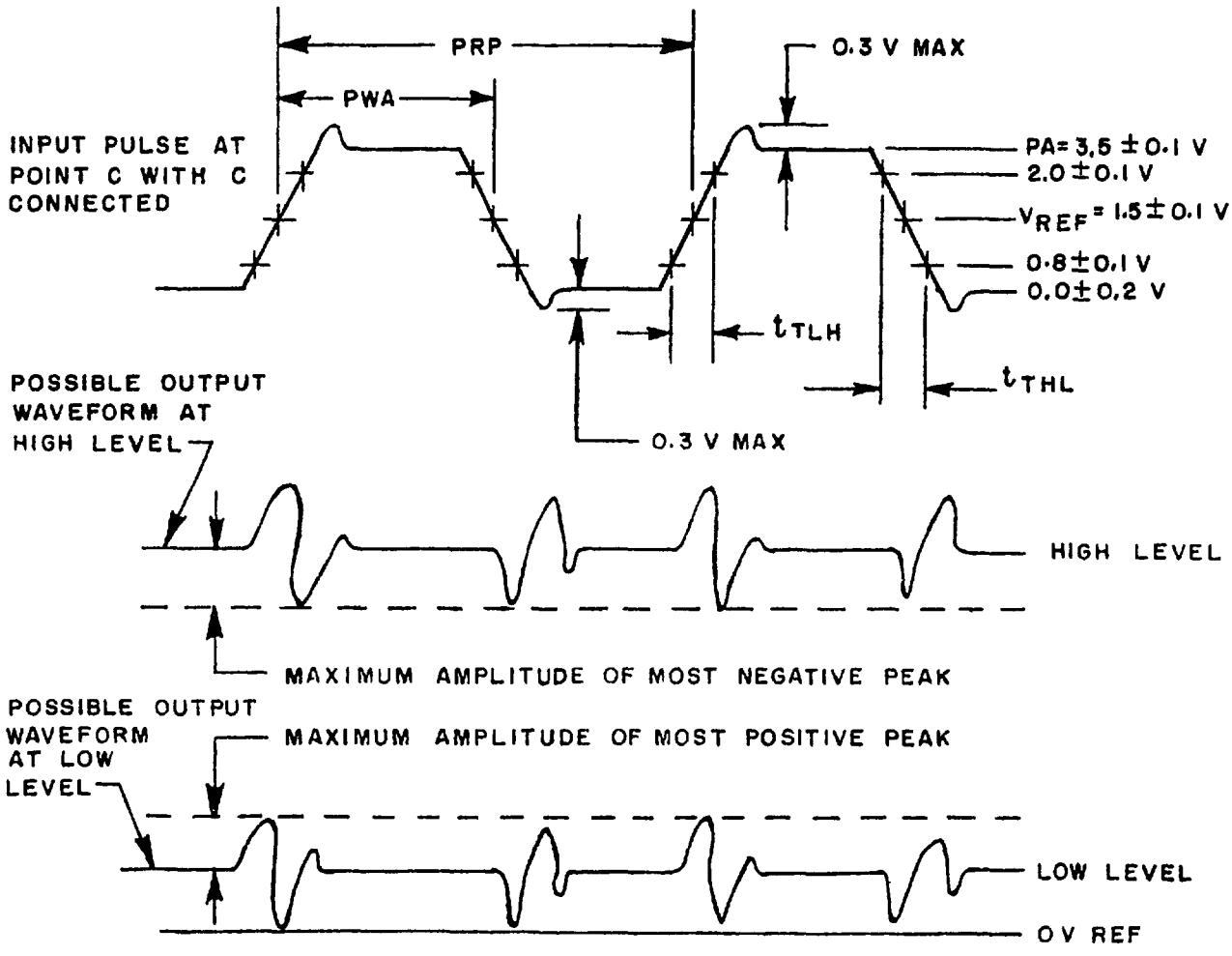
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load all outputs with load as specified on figure 15.
3. R2 = 51.1Ω ±1%, 1/4 W.

FIGURE 16. Output crosstalk (high-level and low-level) test circuit for tables XVI and XVII.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 5.0 \pm 1.5$ ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±10 ns.
2. The lead length of electrical conductors between resistor R2 and input pins, probes Y01-1 and Y01-2 to input and output pins, load circuits to output pins, and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins shall be less than 1 inch (25.4 mm).

FIGURE 17. Output crosstalk (high-level and low-level) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	18,19

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S	Power supply filter capacitance	Open	1	3.4	6.2	µF	20

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TABLE XX. Input capacitance test.

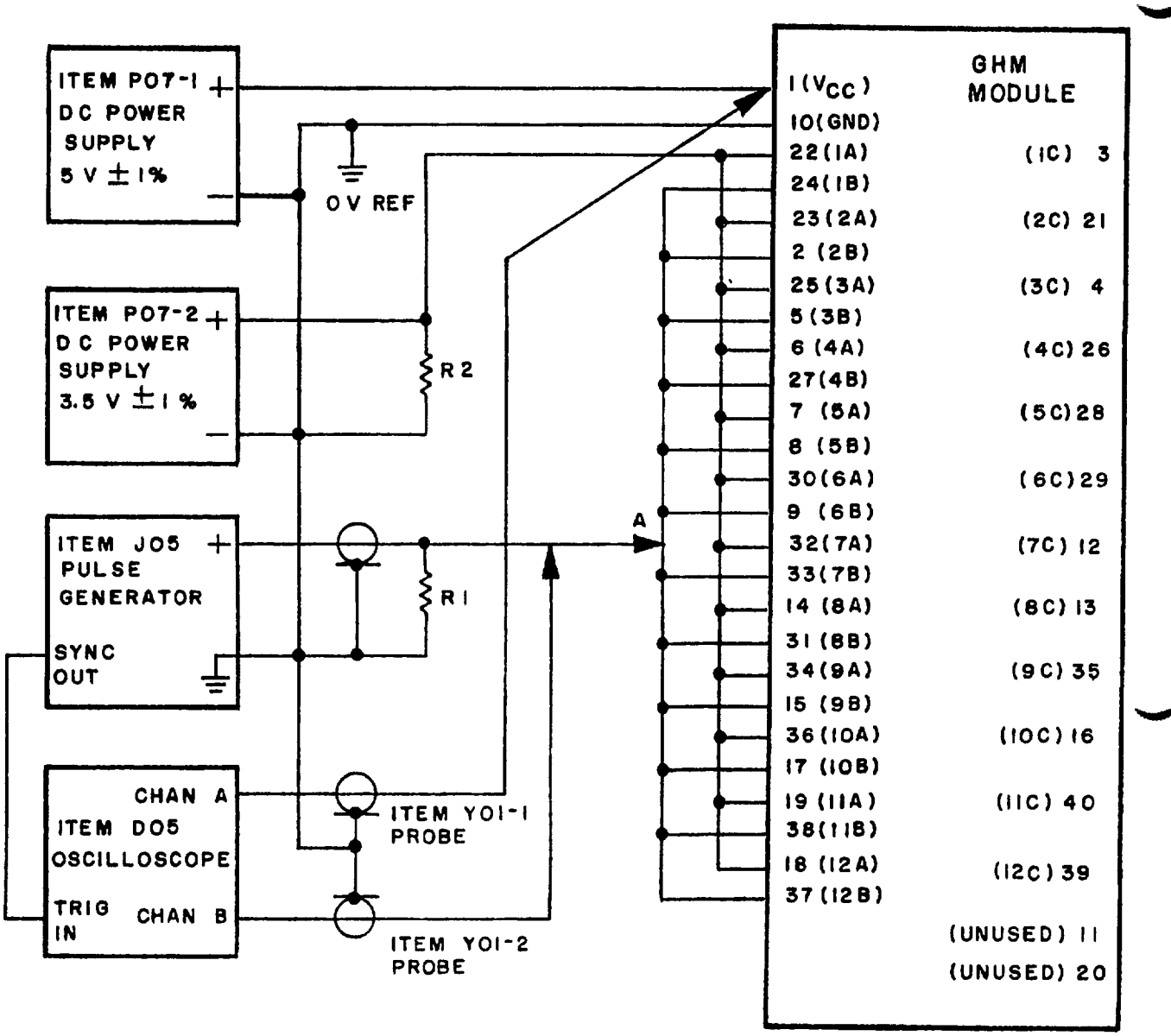
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	1A Input capacitance	22	Open	2	15	pF	20
T2	1B "	24	"	"	"	"	"
T3	2A "	23	"	"	"	"	"
T4	2B "	2	"	"	"	"	"
T5	3A "	25	"	"	"	"	"
T6	3B "	5	"	"	"	"	"
T7	4A "	6	"	"	"	"	"
T8	4B "	27	"	"	"	"	"
T9	5A "	7	"	"	"	"	"
T10	5B "	8	"	"	"	"	"
T11	6A "	30	"	"	"	"	"
T12	6B "	9	"	"	"	"	"
T13	7A "	32	"	"	"	"	"
T14	7B "	33	"	"	"	"	"
T15	8A "	14	"	"	"	"	"
T16	8B "	31	"	"	"	"	"
T17	9A "	34	"	"	"	"	"
T18	9B "	15	"	"	"	"	"
T19	10A "	36	"	"	"	"	"
T20	10B "	17	"	"	"	"	"
T21	11A "	19	"	"	"	"	"
T22	11B "	38	"	"	"	"	"

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T23	12A Input capacitance	18	Open	2	15	pF	20
T24	12B Input capacitance	37	Open	2	15	pF	20

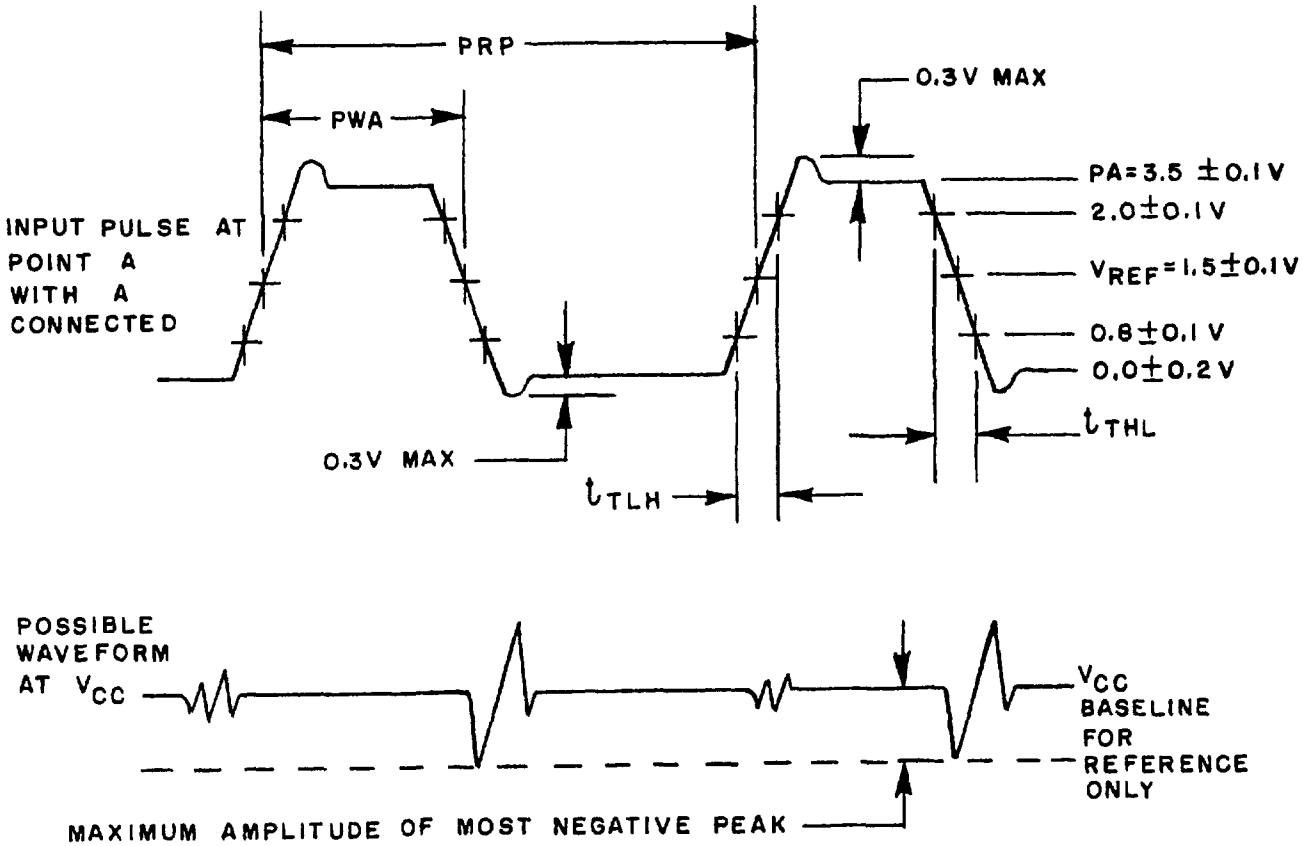
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. R2 = 51Ω ±5%, 1 W.

FIGURE 18. Power supply transient amplitude test circuit for table XVIII.

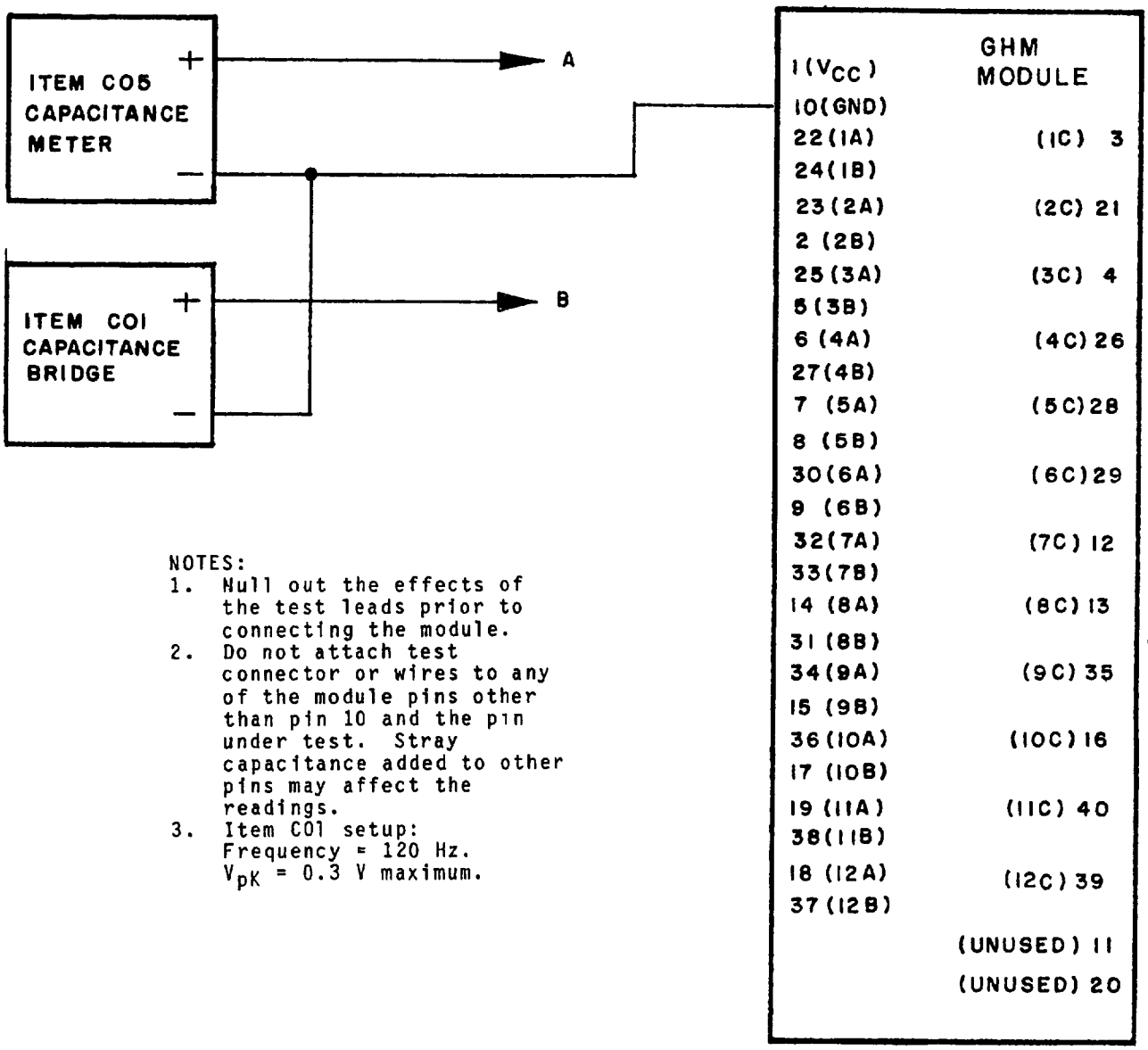
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.

FIGURE 19. Power supply transient amplitude test circuit waveforms for table XVIII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup: Frequency = 120 Hz. V_{pK} = 0.3 V maximum.

FIGURE 20. Power supply filter and input capacitance test circuit for tables XIX and XX.

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